



SARUS AEROSPACE
PRIVATE LIMITED · INDIA

ROADMAP PROPOSAL

Future Plans

Where Sarus Operation Planner and the Sarus firmware could go after version 1.1, and the order we suggest doing it in.



COVERS	BASELINE	DATE	STATUS
App, firmware, quality, business	App 1.1.0 · Firmware Sarus-2 (4.7.1 and 4.6.3)	6 October 2026	Proposal for discussion

01 Starting point

Version 1.1 gives Sarus a ground station of its own and control over who may change an aircraft. It does everything Mission Planner does and loses fewer messages. Changing an aircraft's setup needs the owner's password, enforced by the program and by Sarus firmware, and parameter values are taken exactly as set. Each airframe's physical limits are watched, and an alarm sounds when a mission is not going as planned. The firmware is built automatically on two ArduPilot lines, 4.7.1 and 4.6.3.

All of it has been tested in ArduPilot's simulator, on QuadPlane, Copter and Rover, including a full regression of version 1.1 and an engine failure in flight, and it installs cleanly on fresh Windows 10 and 11 machines. None of it has flown a real aircraft yet. That makes the field test the first item on any list, and it is why the other ideas here are proposals rather than commitments.

3 vehicle types tested in simulation	~3,070 parameters written, read back and restored	0 changes to Mission Planner's keyboard shortcuts; 2 suggestion features removed on request	2 Sarus firmware lines, each for 7 board families
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Each idea below carries a priority. P1 means it is needed to ship and run version 1.1 safely, within weeks. P2 items bring clear value to pilots or customers over the next few releases. P3 marks larger projects that depend on the earlier ones.

02 Finishing version 1.1

A short list stands between the current build and a public release, and all of it is P1.

The field test comes first. Radios, sensors and real airframes behave in ways a simulator does not, and the Field Test Checklist exists for exactly this: bench work with the propellers off, then a staged first flight.

Code signing is the second. Without it, Windows greets every new user with "Windows protected your PC", which is a poor first impression for a product sold with aircraft. It needs a D-U-N-S number for the company and then an organisation-validated certificate with cloud signing.

Two smaller fixes belong here as well. The update package still carries Mission Planner's own change log, and the installer file is named after the Mission Planner base version (1.3.83) instead of 1.1.0. Both should say Sarus.

The owner tries the admin password, and the spare, on a simulator built with the owner keys; nobody else can, because nobody else knows them. Then Sarus-2 is released on both firmware lines and the app as version 1.1.0. Before anything is distributed publicly, a lawyer should review the terms, privacy notice and safety disclaimer, and someone should run a trademark search on "Sarus". Last, both GitHub repositories still live under a personal account and should move to a company organisation.

03 Firmware

The firmware can now carry the owner's parameter set as its defaults: every board build reserves room for it and merges it in, leaving out the values that belong to one aircraft. What is missing is the set itself. A Sarus QuadPlane set would give every new aircraft the same frame settings, output assignments and TECS and VTOL starting values, so setup would be faster and a whole class of configuration mistakes would disappear. P1

The limits question is answered. Of about 3,000 parameters, ArduPilot overrides exactly one for a reason of its own: loiter acceleration, capped at $g \cdot \tan(\text{lean angle})$, which is physics. The others are measured values and 8-bit storage, which cannot change without scrambling every aircraft's stored parameters. Nothing was removed; the program shows what the aircraft keeps, and the airframe limits page warns before a value asks too much. If the flight team finds a limit that truly gets in the way, it gets its own release with simulator evidence from before and after. P3

Two habits would protect all of this. ArduPilot ships its own autotest suites for Plane, Copter and Rover, and running them on every Sarus change would catch most regressions before a release. And each new stable ArduPilot release should be merged into a new Sarus branch within days, because ArduPilot's safety fixes have to reach Sarus aircraft quickly. P2

The parameter lock has one gap: anyone with the board on a bench can flash other firmware over USB. ArduPilot's secure bootloader, installed with a Sarus signing key, would accept only Sarus-signed firmware and close it. P2 Longer term, Sarus could build for every board ArduPilot supports. That is roughly 1,100 firmware files per release, more than GitHub allows in a single release, so it would mean one release per vehicle type. P3

04 Ground station

The rule from version 1.0 stays: Mission Planner's buttons, screens and shortcuts remain where they are, and new functions go next to them. The one exception was made in 1.1 on the owner's request: the two features that wrote suggested parameter values are gone.

For pilots

Text looks slightly soft on screens set to more than 100% scaling, and most new laptops run at 125% or more. Fixing that is the most visible improvement we could make. P2

A QuadPlane pre-flight checklist inside the existing PreFlight tab would put the Sarus procedure where the pilot already looks. Offline map packs, prepared in the office and copied to the field laptops, would remove the grey map that appears wherever there is no internet. A history of parameter changes, recording who changed which value and when, would help tuning and incident reviews, and the lock already knows when a change was unlocked. The airframe limits could grow into full airframe profiles that compare an aircraft with its approved Sarus settings before flight. P2

Further out, a view that plots airspeed, throttle, pitch and height error during transitions would shorten QuadPlane tuning sessions. Hindi and other Indian languages can be added through the translation system the program already has. P3

For regulation in India

Operators have to keep flight records, and a per-flight summary taken from the telemetry log (aircraft, UIN, pilot, times, area and maximum height) would save that paperwork. P2 Showing the Digital Sky green, yellow and red zones on the PLAN and DATA maps depends on what data and terms are available, and permission and Remote ID workflows should wait until DGCA's requirements for Sarus aircraft categories are clear. P3

Security

Version 1.1 locks the aircraft's setup behind the owner's password, in the program and in the firmware. Two steps would complete it. The secure bootloader (above) stops reflashing around the lock. MAVLink message signing, with keys handed out per aircraft and per station, would stop strangers from even sending flight commands over the radio; it needs a procedure for issuing keys, so that it does not also lock out the owner's own pilots. P2

05 Testing

So far the test harness has only ever flown simulated aircraft. A hardware-in-the-loop bench, with a real flight controller on a desk connected to the simulator, would let the same tests exercise real USB connections, real firmware and real timing. If a dedicated Windows machine ran the full regression every night, each change would be checked without anyone having to start it. With the user's consent, crash reports could come back to Sarus, so problems in the field are seen early. P2

Some fixes deserve to go back upstream, the lost-reply fix most of all. Every fix Mission Planner or ArduPilot accepts is code Sarus no longer has to maintain alone. Merging each new Mission Planner release, followed by the full regression, keeps Sarus in step with the original. P3

06 Other platforms

Today the program runs on Windows only. Most of what it does, the communication, parameters, missions and logs, is shared .NET code that runs elsewhere already. The work lies in the interface. We would rebuild it one screen group at a time (DATA, PLAN, SETUP, CONFIG) with a cross-platform toolkit, running the same test harness against each platform, and release nothing until a platform matches the Windows version feature for feature and shortcut for shortcut.

An Android tablet version for field use makes sense first. macOS and Linux versions, mostly for engineering work, can follow.

07 Business and support

Customers will need a place to find the manual, release notes and short how-to videos, so a documentation website comes early. So does a support process: one address, an issue tracker and agreed response times. Releasing the app and firmware on a regular schedule, each with release notes and a test report, makes it easier for customers to plan. P2

Training courses for pilots and technicians can be built from the manual and checklist once those have settled. A fleet view, showing several aircraft and their maintenance state on one screen, becomes useful as the number of Sarus aircraft grows. [P3](#)

08 Suggested order

STEP	WORK	PRIORITY
1	Owner password acceptance test, Sarus-2 and app 1.1.0 releases, field test, code signing, Sarus change log and installer name, the owner's default parameter sets, legal review, company GitHub organisation	P1
2	Secure bootloader with a Sarus signing key; ArduPilot autotests in the firmware build; MAVLink signing procedure	P2
3	Sharp text on high-resolution screens, QuadPlane checklist, parameter history, airframe profiles, offline maps	P2
4	Hardware-in-the-loop bench and nightly regression; message signing procedure	P2
5	Compliance flight-log export, documentation website, support process	P2
6	Android version, all boards, signed firmware, transition monitor, Indian languages, airspace overlay, fleet view	P3

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